DIGITIZING HAND-DRAWN LOGIC CIRCUITS USING COMPUTER VISION

Interim Project Review

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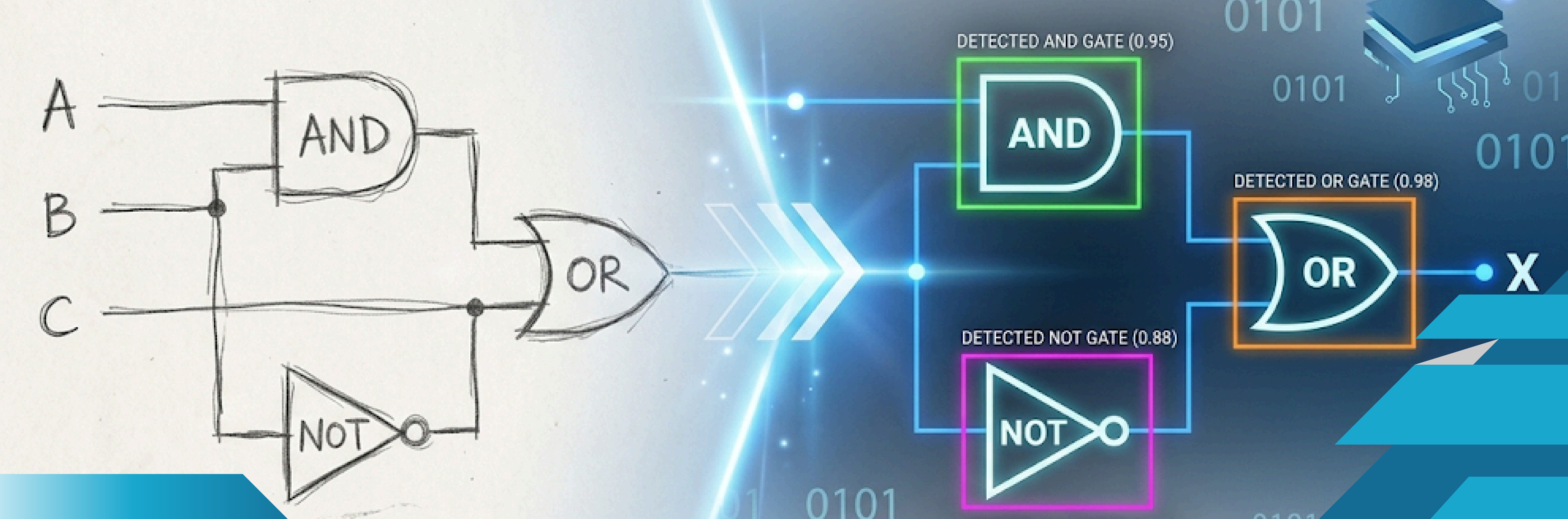
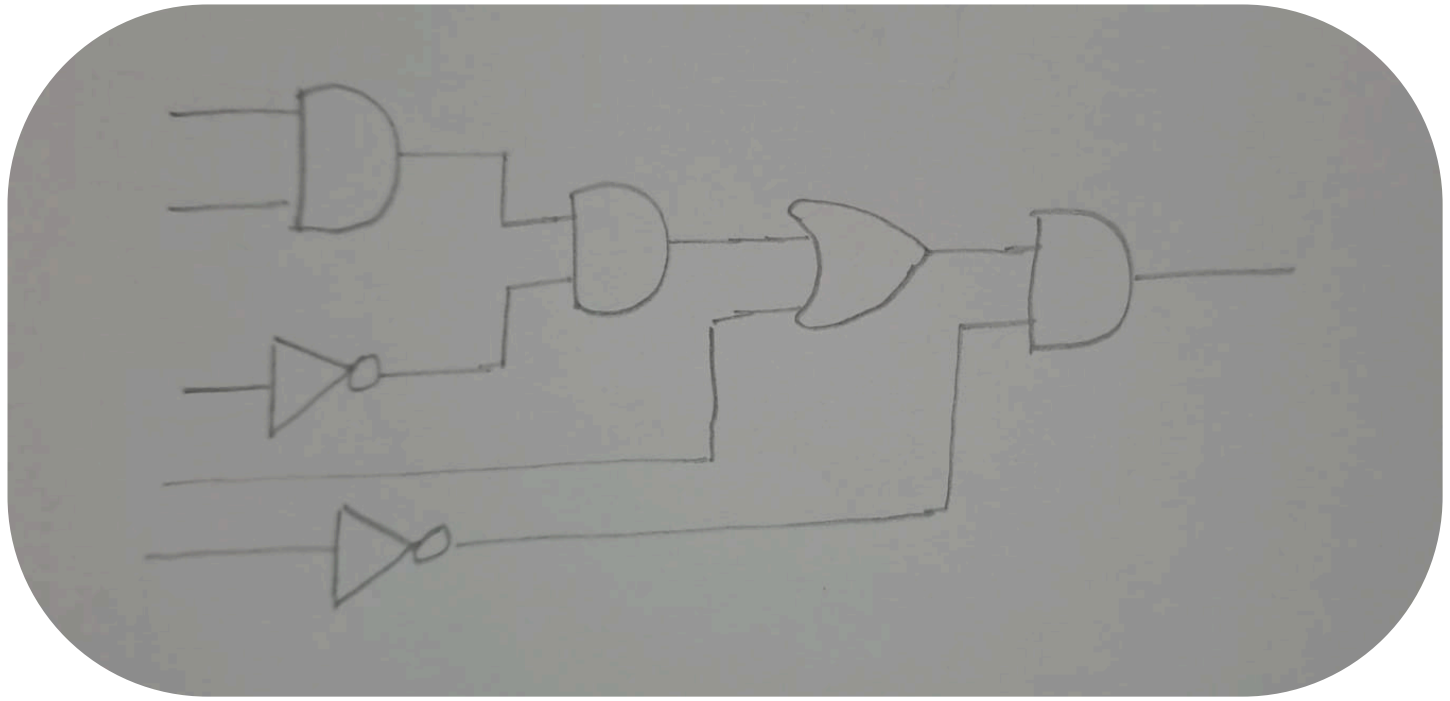
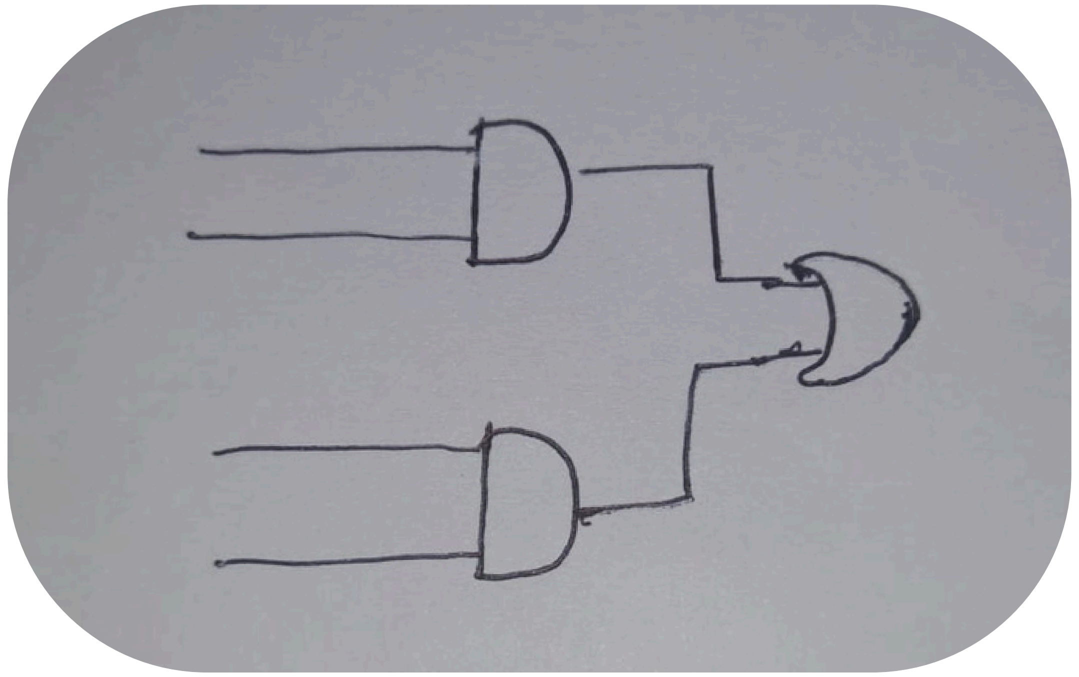


THE PROBLEM:

- Translating inconsistent, hand-drawn logic circuits into functional digital netlists using pure Computer Vision.





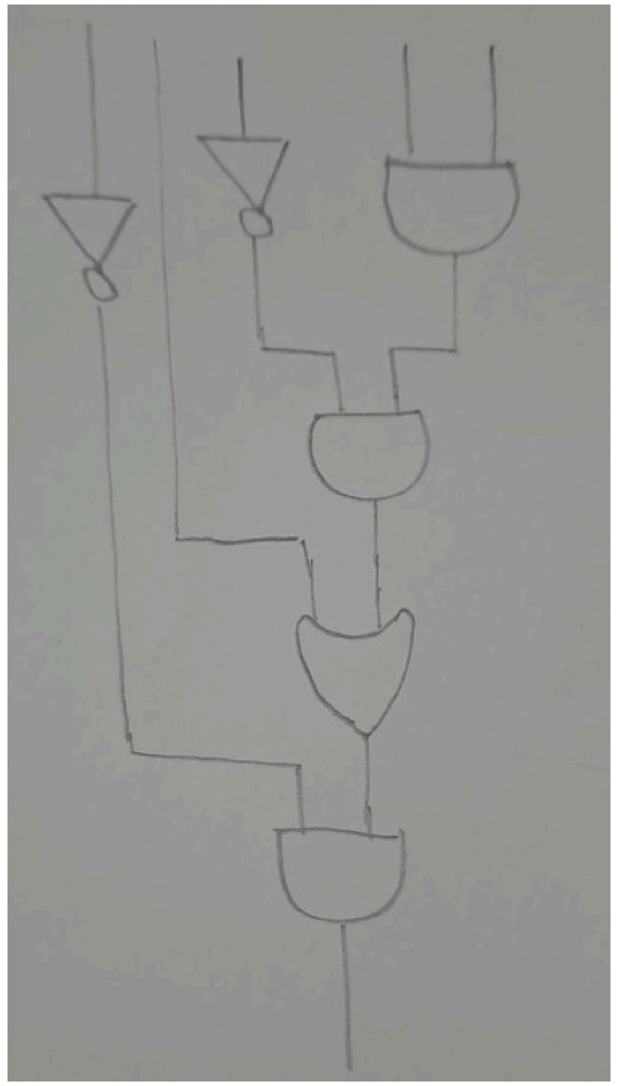
THE CHALLENGE OF HUMAN VARIANCE:

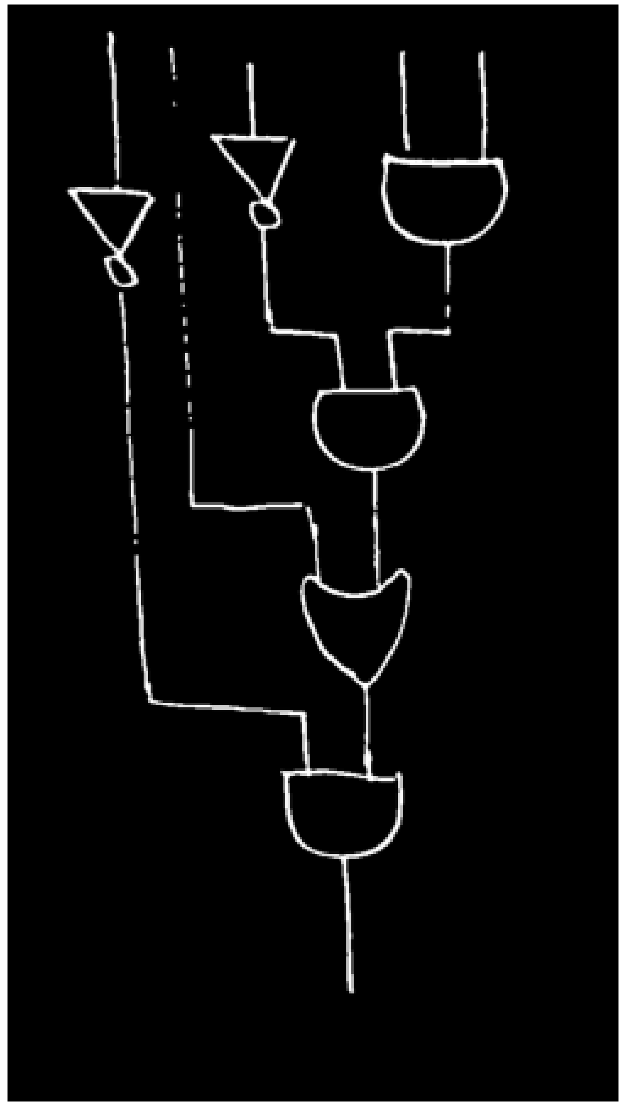
- Heavy ink and overlapping traces.
- Wobbly lines and imperfect geometry.

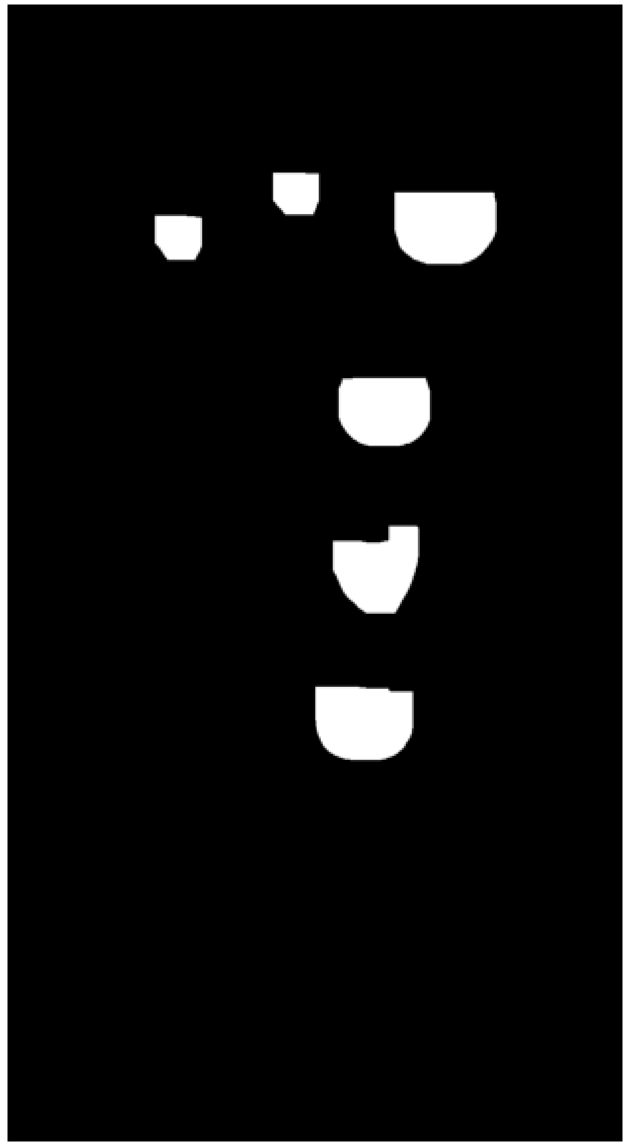
THE DATASET:

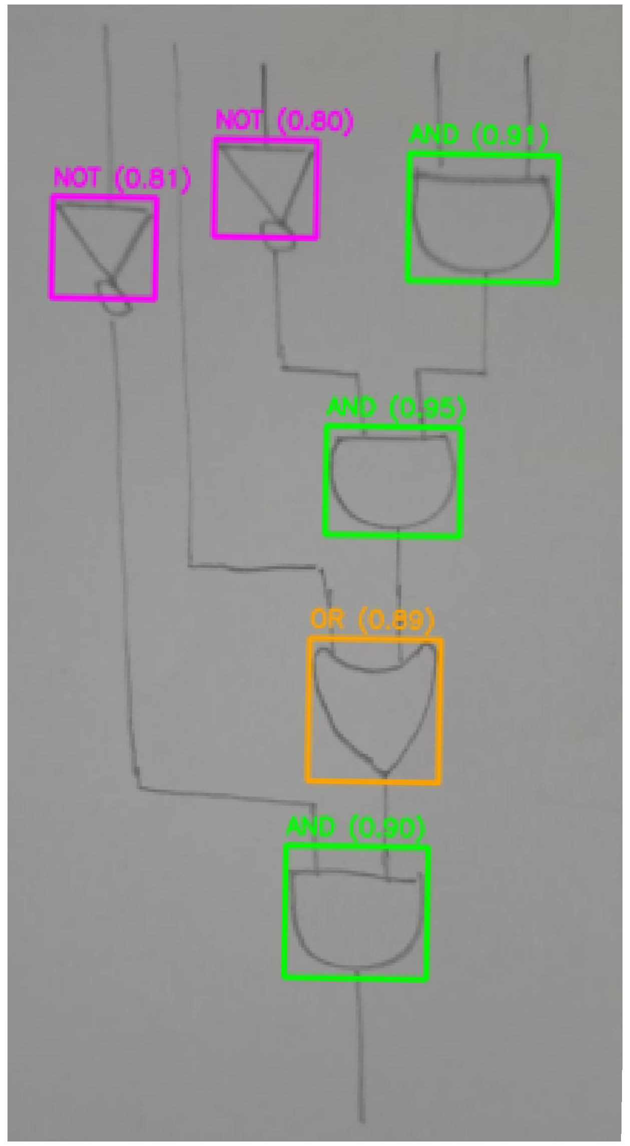
- A custom collection of hand-drawn circuits on paper, scaling from simple 2/3-gate topologies to complex structures.

THE MULTI-STAGE IMAGE PROCESSING PIPELINE









STEP 1: PREPROCESSING

Grayscale conversion, Median Blur, Gaussian Adaptive Thresholding.

STEP 2: GAP SEALING & FLOOD FILL

Morphological closing to handle human variance in pen pressure.

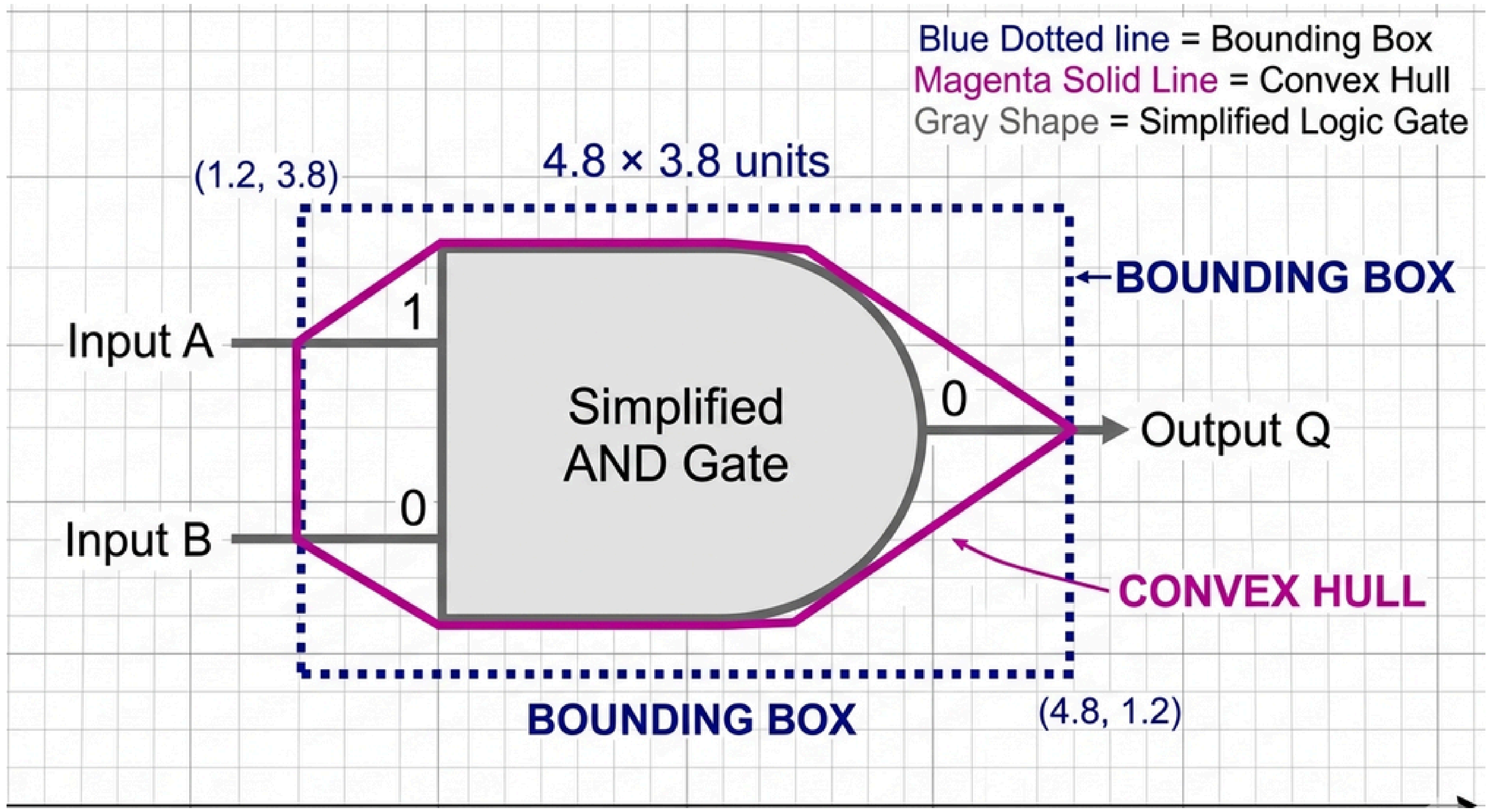
STEP 3: ADVANCED SEGMENTATION

"Jaws of Life" Heavy-Duty Morphological Opening (35x35 kernel).

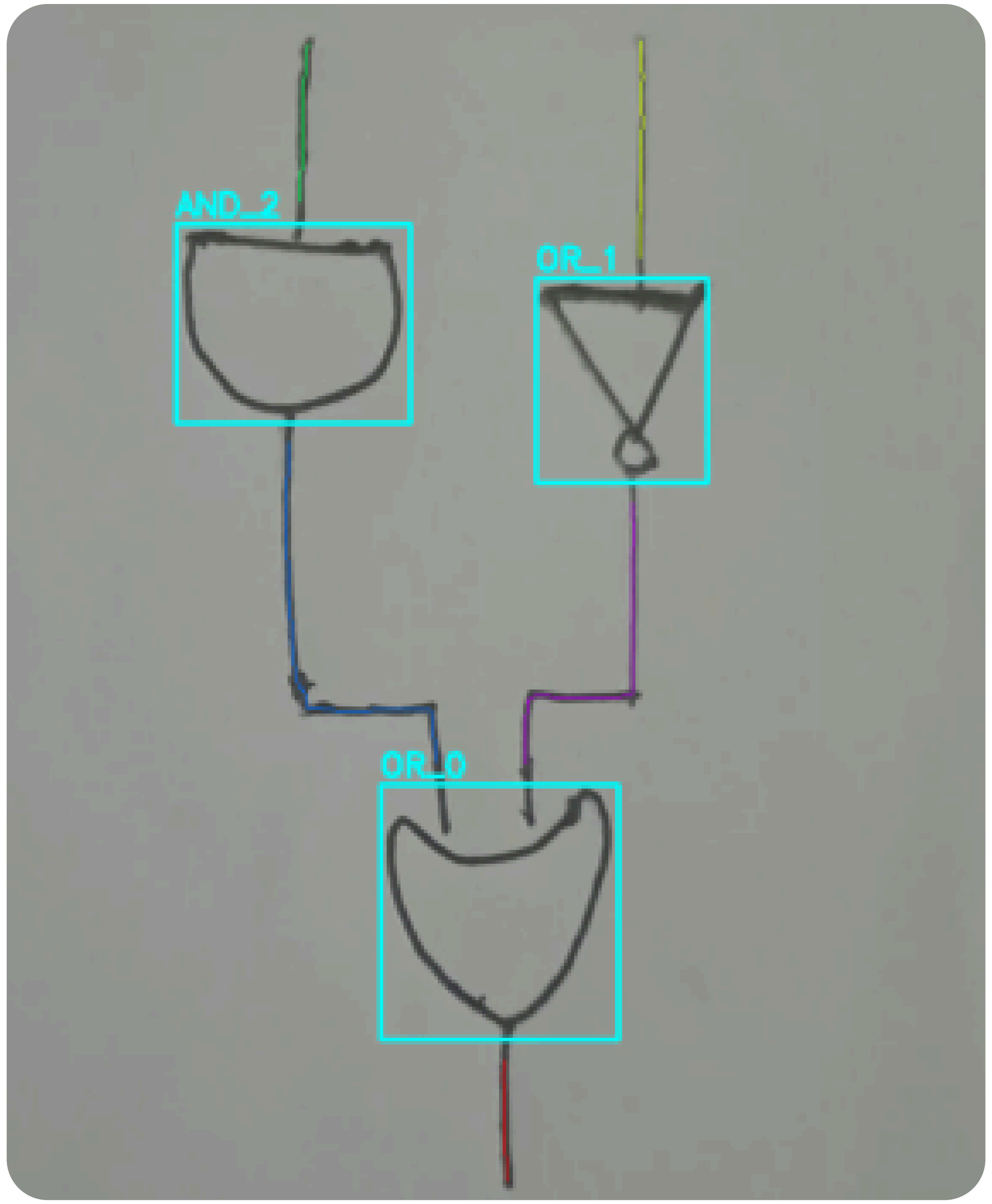
DETERMINISTIC HEURISTIC CLASSIFICATION

Utilizing deterministic pure geometry rather than probabilistic machine learning models.

- **Metric 1: Solidity:** (Area / Convex Hull Area). Detects the complex "dent" in the back curve of an OR gate versus the solid block of an AND gate.
- **Metric 2: Extent:** (Area / Bounding Box Area). The ultimate "Triangle Trap." A triangle mathematically cannot fill its bounding box, instantly isolating NOT gates regardless of size.



- **Generalization Success:** High robustness against human variance in geometry, wobbly lines, and sizing.



FROM PIXELS TO LOGIC

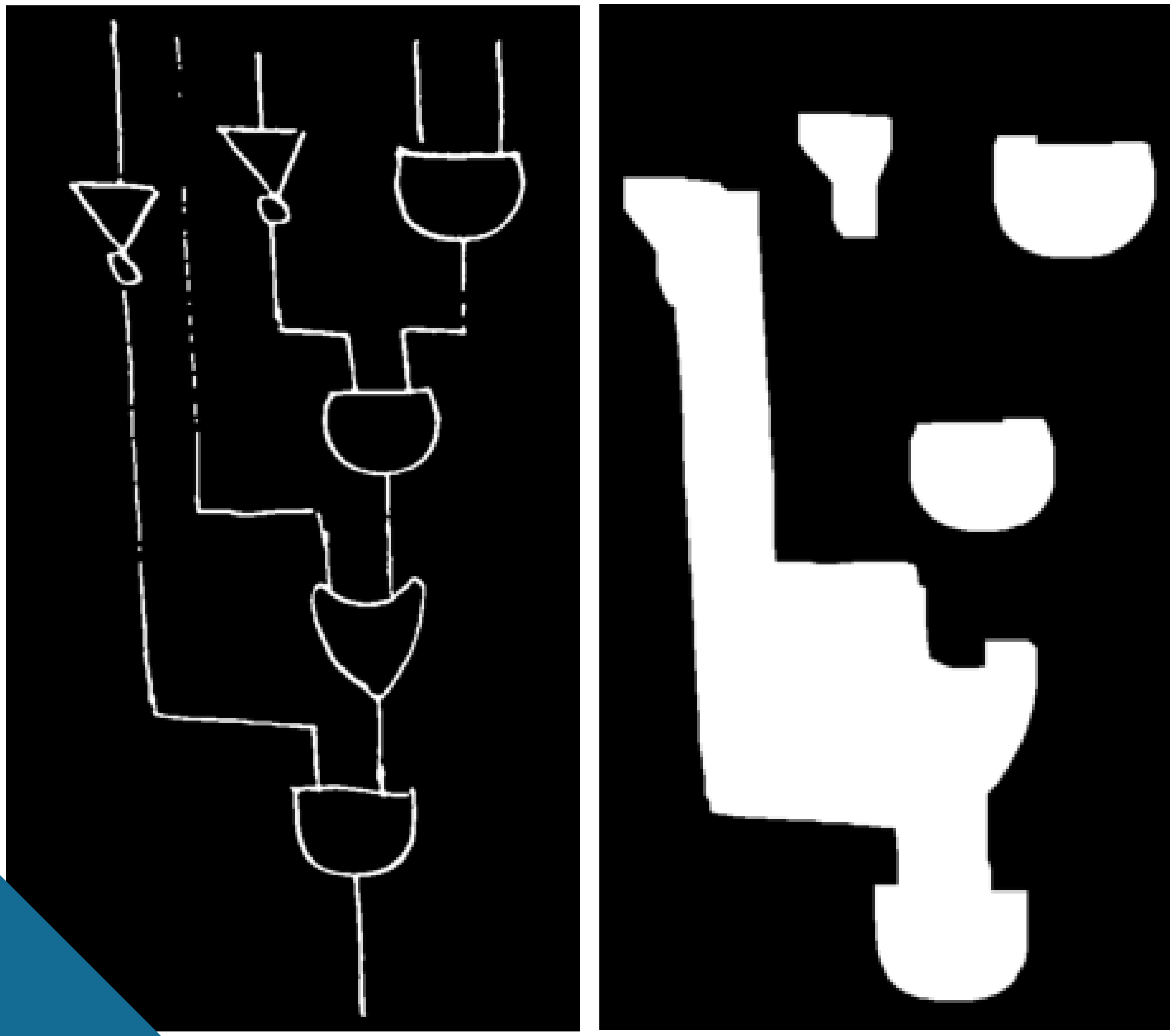
NETLIST GENERATION

- **Gate Masking:** Erasing detected bounding boxes to isolate the raw, unconnected wire pixels.
- **Skeletonization:** Applying morphological thinning to compress thick, wobbly hand-drawn traces into precise 1-pixel-wide paths.
- **Graph Pathfinding:** Treating gates as 'Nodes' and skeletonized wires as 'Edges' to map connections and export a structured JSON Netlist.

LOGIC EXECUTION

- Parsing the JSON Netlist to virtually rebuild the circuit geometry.
- Applying standard Boolean logic to automatically calculate and output the final Truth Table.

CURRENT STATUS : A fully functional, end-to-end pipeline from physical paper to digital simulation.



TECHNICAL CHALLENGES

- **The Segmentation vs. Integrity Paradox:** Weak morphological operations left interconnecting wires, creating "Giant Blobs." Stronger operations cleanly severed wires but "decapitated" tiny NOT gates, destroying their geometric signature.
    - **Solution:** Developed a localized, surgical "Goldilocks scalpel" (13x13 elliptical opening) applied strictly within padded bounding boxes.
- **Brittle Thresholding (Overfitting):** Static limits (e.g., Solidity > 0.85 or specific corner counts) failed when human variance produced borderline geometry (e.g., a wobbly line registering as 8 corners).
    - **Solution:** Pivoted to a multi-factored, deterministic decision tree where Extent acts as the ultimate "Triangle Tie-Breaker" rule.

FUTURE SCOPE & REMAINING WORK

NEXT PHASE 1:

Expanding the Combinational Library

- Extending the geometric classifier to identify complex derived gates: NAND, NOR, XOR, and XNOR.
- Tuning the pipeline to handle extremely dense, tightly packed circuit diagrams.

NEXT PHASE 2:

The Sequential Logic Leap

- Moving beyond feed-forward combinational logic to detect circuits with state and memory.
- Identifying feedback loops and sequential elements like Latches and Flip-Flops (D, T, SR, JK).

THANK YOU